Case Application: Virgin Australia

From One Implied Volatility to the Full Brent Volatility Smile

FNCE30007 Derivative Securities

Learning Objectives

- Use the BSM formula for options on futures with real Brent crude oil market data.
- Understand how implied volatility is inferred from one observed option price.
- Test the constant-volatility assumption by applying one anchor implied volatility across different strikes.
- Interpret the full cross-strike implied-volatility smile, including both the lower-strike and higher-strike wings.

1. Case Background

Fuel price volatility is a major source of risk for airlines because jet fuel represents a significant share of operating costs. Virgin Australia actively manages this exposure through a fuel-hedging program involving Brent crude oil, refining margins and derivative contracts.

During the 2026 fuel-price shock, Virgin reported that a substantial proportion of its Brent crude oil and refining-margin exposure had already been hedged. The experience of Qantas during the same period also highlighted how large changes in fuel prices and refining margins can materially affect airline operating costs.

Virgin's individual OTC option contracts are not publicly disclosed in sufficient detail to observe their exact premiums and contractual terms. Therefore, this case uses exchange-traded ICE Brent crude oil futures options as a transparent market proxy.

Different strike prices represent different levels of fuel-price protection. We first use one approximately at-the-money Brent call to obtain a market-implied volatility. We then ask whether that single volatility can explain prices across a much wider range of strikes. Finally, we recover the individual implied volatilities across the option chain and examine the complete volatility smile.

Market option data are taken from a Barchart snapshot of Crude Oil Brent Dec '26 (CBZ26), viewed 17 September 2026. Bid and ask quotes are used to form market mid-prices.

2. Market Setup

For the December 2026 ICE Brent crude oil futures options, use:

$F_0 = 101.10$

$T = 41/365$

$r = 4.00\%$ p.a. (assumed for simplicity)

Market Mid-Price = (Bid + Ask) / 2

To construct a cleaner cross-section of the smile, use out-of-the-money (OTM) puts for strikes below the futures price and OTM calls for strikes above the futures price. This avoids relying on deep in-the-money quotes on either wing and is consistent with common option-smile construction practice.

3. BSM Formula for Options on Futures

For calls:

$C = e^{(-rT)} [F_0 N(d_1) - K N(d_2)]$

For puts:

$P = e^{(-rT)} [K N(-d_2) - F_0 N(-d_1)]$
$d_1 = [\ln(F_0/K) + 0.5\sigma^2 T] / (\sigma\sqrt{T})$
$d_2 = d_1 - \sigma\sqrt{T}$

4. Lecture Application - Where Does Implied Volatility Come From?

Use the approximately at-the-money 101-strike call as the anchor option:

Option	Strike K	Bid	Ask	Market Mid-Price
101C (anchor)	101.00	7.01	7.12	7.065

(a) Set up the BSM futures-call formula using $F_0 = 101.10$, $K = 101$, $T = 41/365$ and $r = 4.00\%$.

(b) Use Excel Goal Seek or Solver to find the value of σ that makes the BSM call value equal to the observed market mid-price of 7.065.

(c) Explain why the resulting σ is called implied volatility rather than directly observed volatility.

(d) If this anchor implied volatility is used for every other strike, what BSM assumption are we imposing?

Suggested Answers

(a) BSM Setup

Substituting the known inputs:

$$C(\sigma) = e^{(-0.04 \times 41/365)} [101.10N(d_1) - 101N(d_2)]$$
$$d_1 = [\ln(101.10/101) + 0.5\sigma^2(41/365)] / [\sigma\sqrt{(41/365)}]$$
$$d_2 = d_1 - \sigma\sqrt{(41/365)}$$

The unknown input is σ and the target option value is 7.065.

(b) Implied Volatility

Goal Seek	Entry
Set cell	BSM call-value cell
To value	7.065
By changing cell	Volatility, σ

Answer: $\sigma_{anchor} \approx 52.22\%$

At $\sigma \approx 52.22\%$: $d_1 \approx 0.0932$, $d_2 \approx -0.0819$ and the BSM call value ≈ 7.065.

(c) Why Is It Called Implied Volatility?

The option price is directly observed in the market, but σ is not. Implied volatility is the value of σ inferred by working backwards through the BSM model so that the model price matches the observed market price.

(d) What Assumption Is Imposed Across Strikes?

Using the same implied volatility for calls with different strikes imposes the BSM constant-volatility assumption: volatility is treated as the same across strikes for the same underlying and maturity.